

Production Line Bottleneck & Efficiency Analysis - Single Simulation Summary Report

1. Executive Summary

- Critical Bottleneck Identified:** A severe, cascading bottleneck exists at the "漫膠區" (Coating Area), with a current cycle time of 595 seconds, over 20x its predecessor's time, crippling the entire line's flow.
- Unstable Output with Hidden Potential:** Hourly production fluctuates significantly (1010 to 1380 units), indicating poor line balancing and control, yet peak output demonstrates the line's latent capacity.
- Inefficient Space Allocation:** The factory layout is asset-heavy, with 40% of space dedicated to equipment, suggesting potential over-investment in machinery versus workflow optimization (AGV paths at 15%, material storage at 25%).

2. Simulation Background & Core Objectives

- Simulation Context:** Analysis of a multi-station production line involving SMT-C, Coating, Testing, and Panel Separation processes, with material handling via AGVs.
- Core Objectives:** Identify primary bottlenecks, analyze line balancing and stability, and evaluate factory layout efficiency to maximize throughput and return on assets.

3. Core Results & Deep Dive Analysis

3.1 Overall Production Metrics

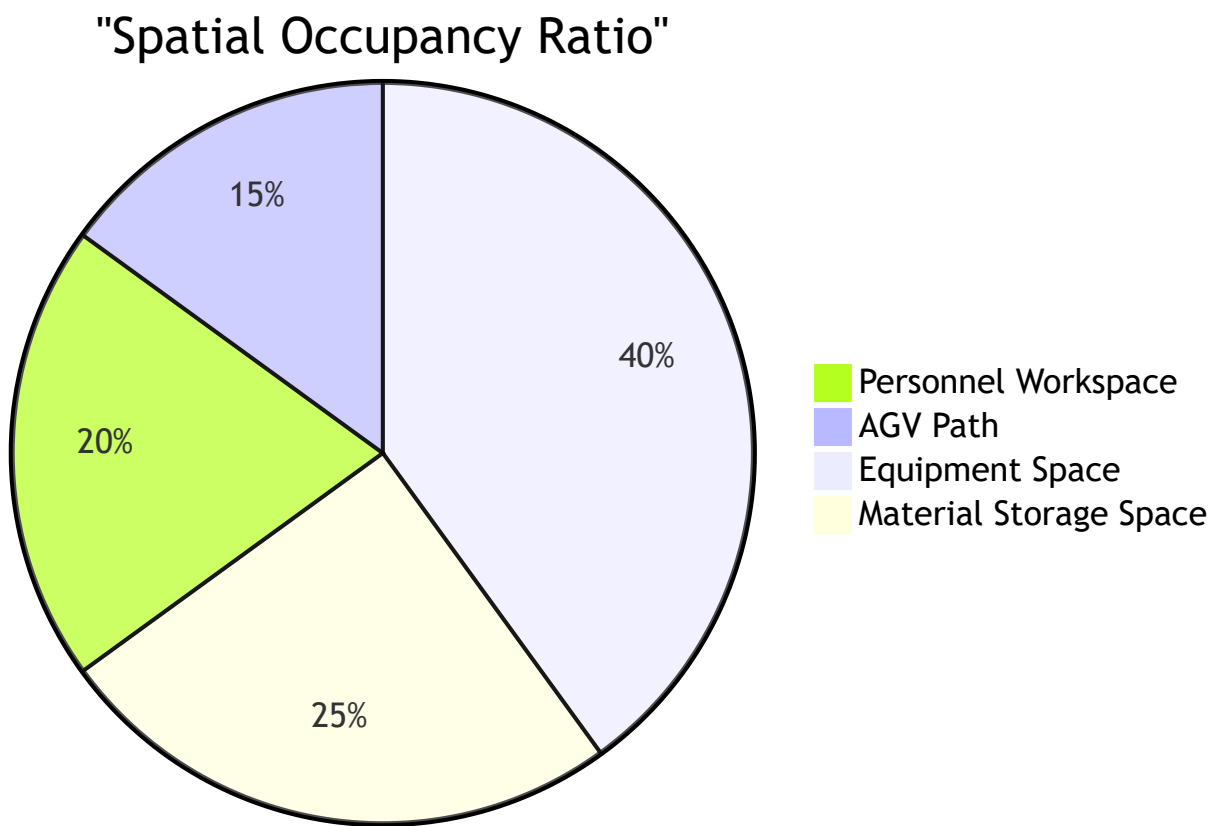
Panel: 700 PCS: 700 Average Takt Time: 187

[Detailed Report]

- Micro-Observation:** The data confirms a 1:1 ratio of Panels to Pieces (PCS), indicating a single-piece flow model for this 700-unit batch. However, the average takt time of 187 seconds is alarmingly high, suggesting a fundamental constraint in the line's speed.
- Comparative Logic:** While the single-piece flow model is theoretically lean, the high average takt time of 187 seconds creates a massive contradiction, revealing a system where flow principles are undermined by severe operational inefficiencies.

- 3. Business Impact:** This slow takt time directly caps theoretical maximum daily output. At 187 seconds per unit, the line cannot exceed approximately 462 units in a 24-hour period, creating a severe ceiling on revenue potential and asset utilization (ROA).
- 4. Operational Deduction:** Root cause: The high average takt is a symptom, not the disease. It is the mathematical result of a dominant bottleneck station dictating the pace for the entire line, as will be proven in subsequent station-level analysis.
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3.2 Spatial Occupancy Ratio



[Detailed Report]

- 1. Micro-Observation:** The factory layout exhibits a pronounced 40% bias towards equipment space, the single largest category, indicating a capital-intensive, machine-centric design philosophy.
- 2. Comparative Logic:** While 40% is allocated to static equipment, only 15% is dedicated to dynamic AGV paths. This 2.7:1 ratio suggests a potential misalignment where the infrastructure for material movement is under-provisioned relative to the processing assets it must serve.
- 3. Business Impact:** This layout inefficiency directly erodes Return on Floor Space (ROFS). High equipment density without proportional flow space increases congestion risk, lowers

flexibility, and can lead to underutilized expensive machinery waiting for transport, compounding capital costs.

4. Operational Deduction: Root cause: A "Push-System" layout philosophy. Space is allocated based on equipment footprint ("just-in-case") rather than optimized for smooth material flow ("just-in-time"). The high material storage (25%) further confirms buffer-driven, rather than flow-driven, operations.

3.3 Station Takt Time Analysis

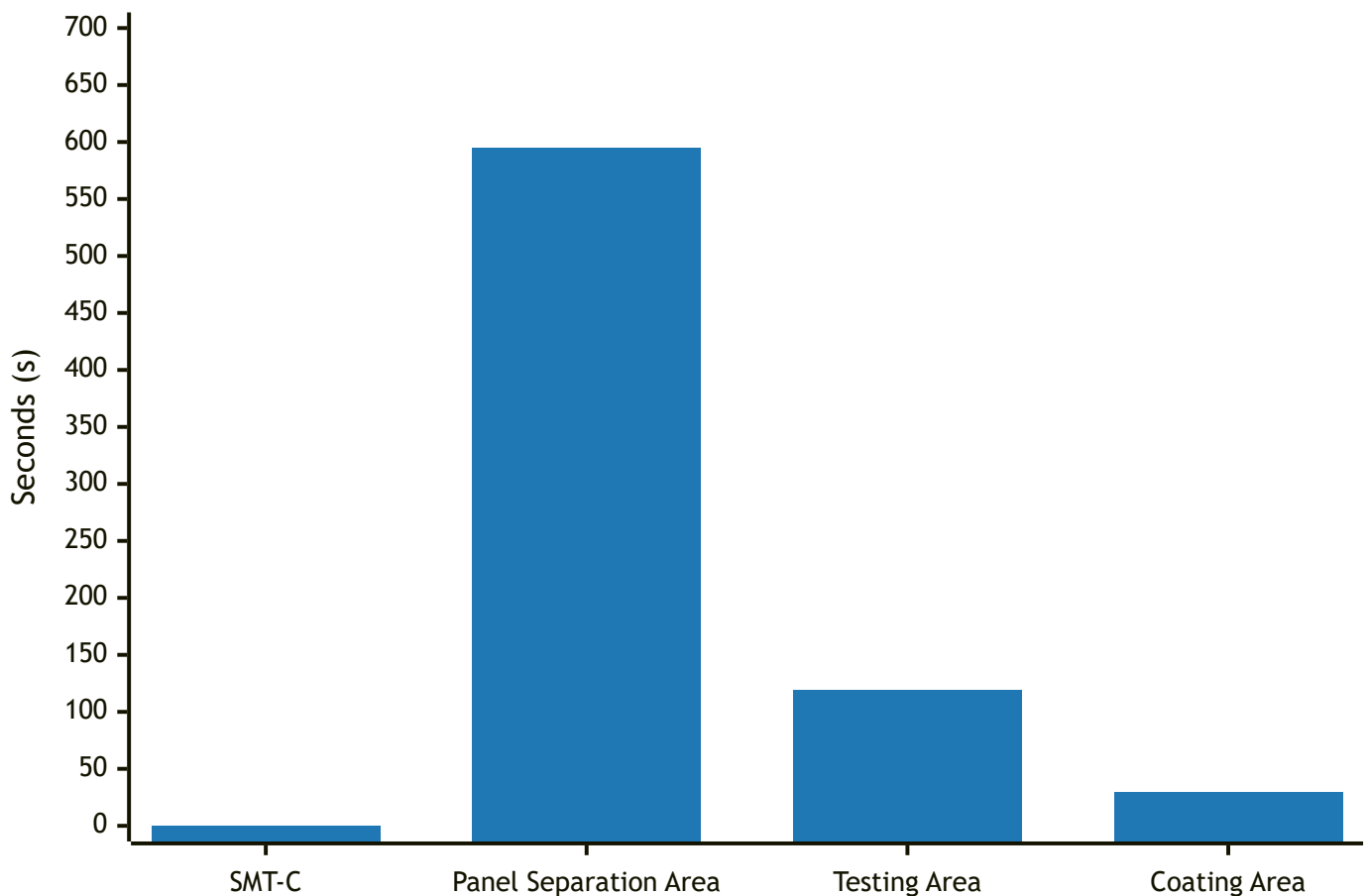
Station	Predecessor Takt	Current Takt
Coating Area	29	595
Testing Area	119	5
Panel Separation Area	595	119
SMT-C	0	29

[Detailed Report]

- 1. Micro-Observation:** The "Coating Area" shows a catastrophic 1,951% increase in cycle time (29s to 595s), transforming it from the fastest to the slowest station. Conversely, the "Testing Area" shows a 95.8% decrease (119s to 5s).
- 2. Comparative Logic:** While the Testing Area's drastic speed-up suggests process improvement or underloading, the Coating Area's explosion creates a system-wide contradiction: downstream stations (Panel Sep, SMT-C) are starved, while upstream (Testing) is blocked, causing a full line seizure.
- 3. Business Impact:** This bottleneck creates a massive "Hidden Factory" of WIP (Work in Progress) waiting at the Coating Area. The direct cost is the tied-up capital in stalled inventory and the lost opportunity cost of the idling downstream capacity, devastating overall equipment effectiveness (OEE).
- 4. Operational Deduction:** Root cause: A **single-point process failure** at the Coating Area. This could be due to a machine breakdown, a drastic increase in manual operation time, a change in material requiring longer cure time, or a severe shortage of operators/AGVs serving this station.
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3.4 Predecessor Takt Time by Station

Predecessor Takt Time by Station

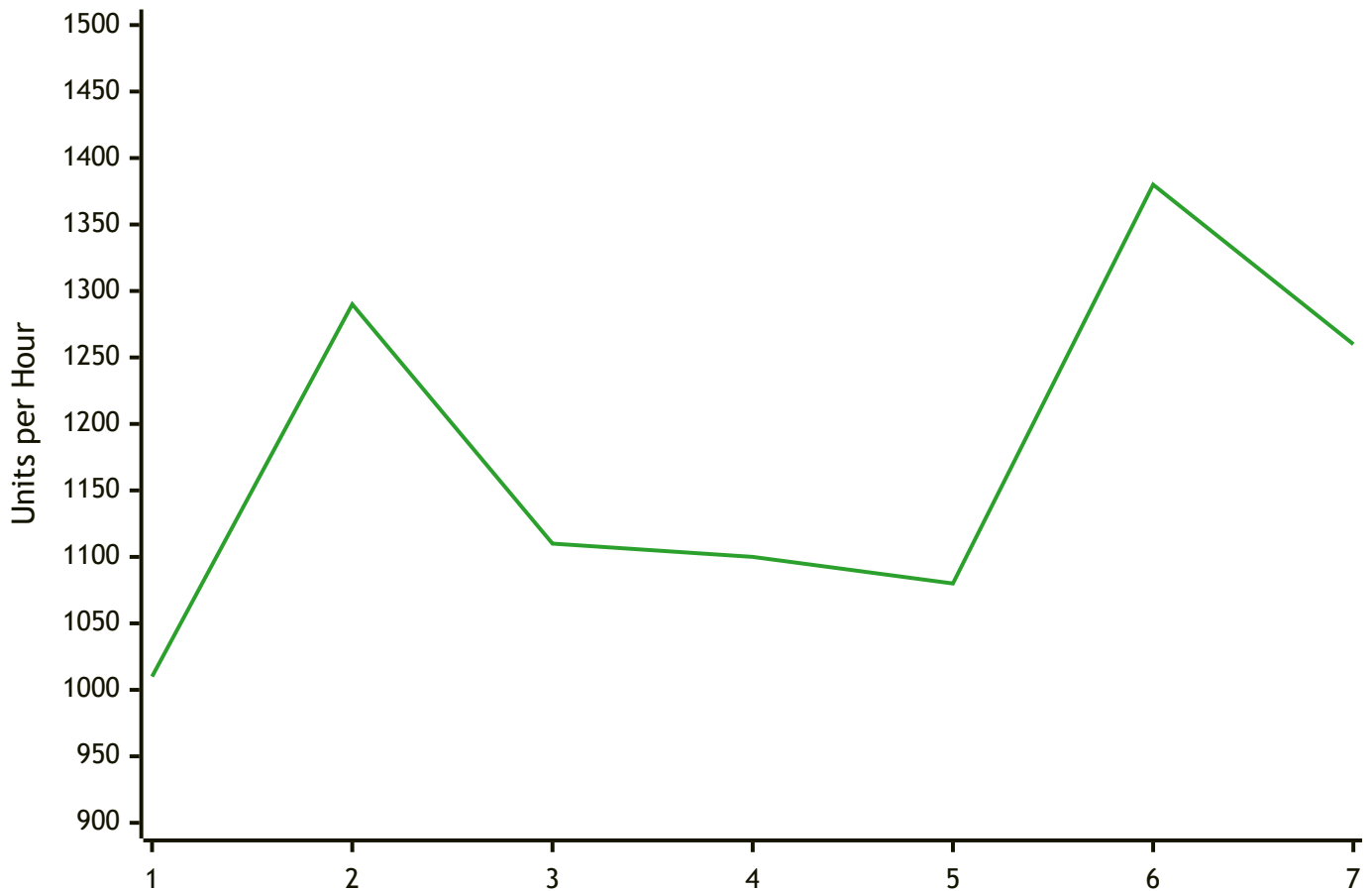


[Detailed Report]

- 1. Micro-Observation:** The predecessor state shows an extreme imbalance, with the "Panel Separation Area" at 595s as the clear bottleneck, and "SMT-C" at 0s indicating it was the line's entry point with no prior constraint.
- 2. Comparative Logic:** The previous state's bottleneck (Panel Sep @ 595s) and the new state's bottleneck (Coating @ 595s) are identical in magnitude but different in location. This indicates the line's weakness shifted but was not resolved; the system's maximum pace remains shackled to a 595-second station.
- 3. Business Impact:** This "bottleneck migration" reveals a reactive, fire-fighting operational culture. Resources were likely diverted to fix Panel Separation, inadvertently starving or overloading the Coating Area, thus moving the constraint without improving overall throughput—a zero-sum game for productivity.
- 4. Operational Deduction:** Root cause: **Localized optimization.** The effort to speed up Panel Separation (from 595s to 119s) was not synchronized with a holistic line re-balancing. The Coating Area, previously fast, became the new constraint due to increased demand or neglected maintenance, proving Goldratt's Theory of Constraints.

3.5 Hourly Output

Hourly Output



[Detailed Report]

1. Micro-Observation: Hourly production shows high volatility, with a range of 370 units (1010 to 1380) and no stable trend. A peak occurs at hour 6, but output in hours 3-5 shows a concerning dip.

2. Comparative Logic: While the peak output of 1380 units/hour demonstrates the line's theoretical capacity, the average output (~1176 units/hour) is 15% lower. This gap between peak and sustained performance indicates unreliable material flow, operator variability, or machine intermittency.

3. Business Impact: This instability makes production planning and delivery commitments hazardous. It erodes customer trust and increases safety stock requirements across the supply chain. The unpredictable output directly translates to unpredictable cash flow and revenue recognition.

4. Operational Deduction: Root cause: **Insufficient buffering and recovery logic.** The line lacks the mechanisms (physical or algorithmic) to smooth out disruptions. The output pattern suggests the bottleneck's severity varies (e.g., Coating Area may have intermittent breakdowns or batch-style processing), causing ripples through the entire system.

4. Strategic Recommendations

Bottleneck Diagnosis

The primary and absolute bottleneck is the **Coating Area** with a 595-second cycle time. This is a "Capacity Constraint" rather than a "Market Constraint," meaning internal process failure is limiting output well below market demand or line potential. The secondary issue is the **highly unstable line balance**, evidenced by the migrating bottleneck and volatile hourly output.

Action Plan

1. Immediate Takt Time Reduction at Coating Area (Week 1-2):

- **Root Cause Analysis (RCA):** Execute a 5-Why analysis on the 595s cycle time. Is it machine speed, manual operation, setup time, or material wait?
- **Kaizen Event:** Mobilize a cross-functional team to address the root cause. Potential actions: machine maintenance/upgrade, operator re-training, work-cell redesign, or introduction of semi-automation.
- **Target:** Reduce Coating Area takt time to $\leq 150s$.

2. Line Re-Balancing & Buffer Strategy (Week 3-4):

- **Re-calculate Ideal Takt:** Based on demand and new Coating capacity, re-balance work across all stations.
- **Implement Strategic Buffers:** Place a controlled WIP buffer *before* the Coating Area to decouple it from upstream variability and ensure it never starves. This protects the constraint's utilization.
- **Implement Pull Signals:** Install a Kanban or CONWIP system to regulate release into the line based on Coating Area consumption, preventing overproduction upstream.

3. Layout Optimization for Flow (Month 2):

- **Redesign for Flow:** Re-evaluate the 40% equipment space allocation. Can any equipment be rearranged or miniaturized to widen AGV paths (15%) and reduce travel time?
- **Point-of-Use Storage:** Reduce the 25% bulk storage by implementing point-of-use material racks along the line, fed by AGVs.

Supplementary Recommendations

- **Data Monitoring:** Implement real-time Andon boards showing station cycle times versus target. Alert immediately if any station exceeds 120% of its takt.
- **Simulation Validation:** Before physically rearranging the layout, use digital twin simulation to test new configurations for AGV traffic and workflow.
- **Focus on OEE:** Shift management focus from individual machine utilization to Overall Equipment Effectiveness for the entire line, prioritizing throughput at the constraint.